

A large dark blue triangle is positioned on the left side of the slide, pointing towards the top-left corner.

My Azure Environment

Name: Mthunzelwa Makwanyane

Student number: 20241063

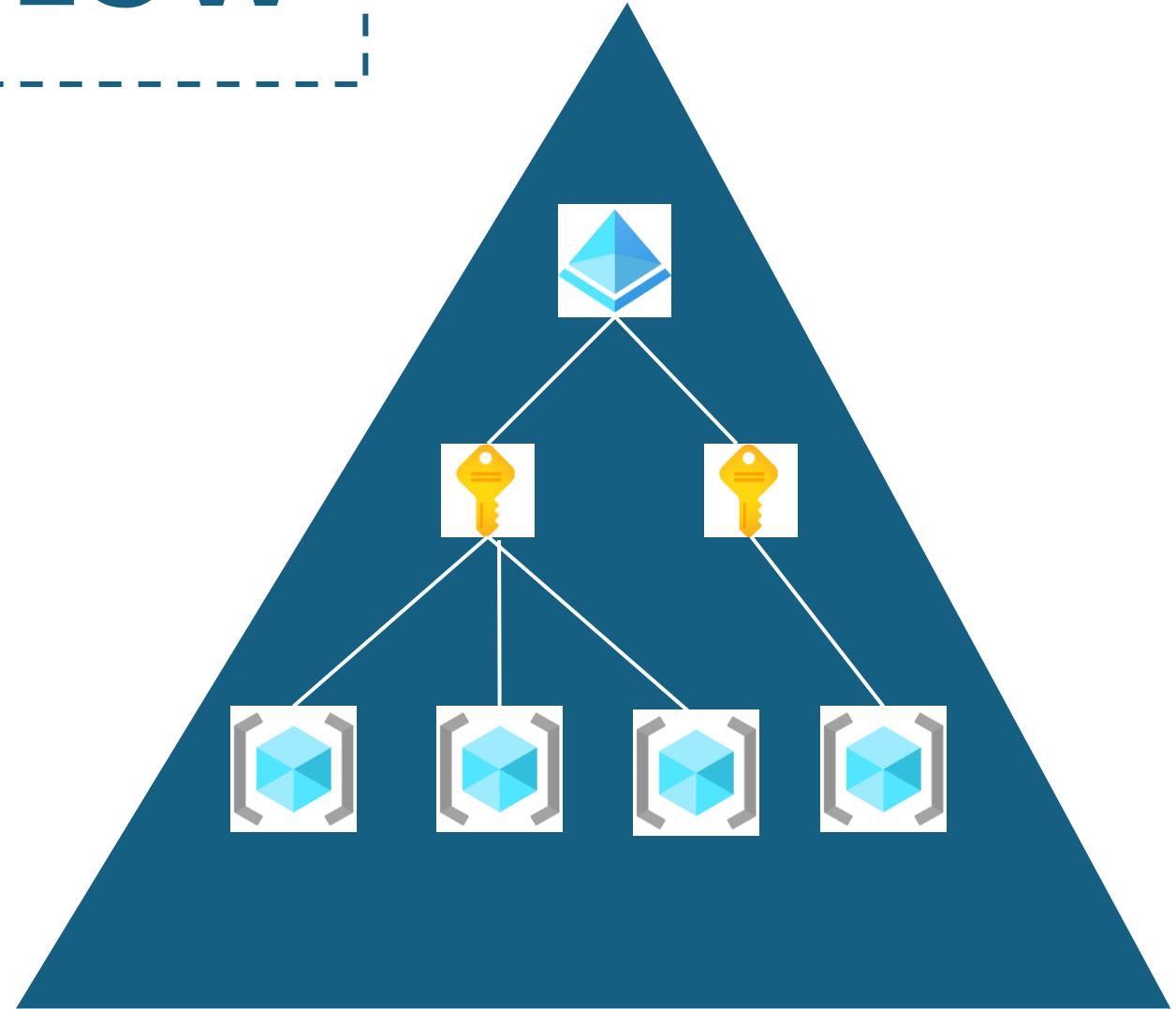
Module: NET 731

Date: 2026/05/10

Project Overview

In this project I basically build an small azure environment designed for the use of a virtual machine with multiple virtual networks including even more NSGs. Within these NSGs and virtual networks, there are diagnostics tools deployed on a couple of these resource to monitor their activity.

This environment serves its purpose of a general simple deployment with ideal resources deployed along with security and monitoring features with only 4 users, each with their specific roles and permissions.



Diagram

To briefly explain my environment shown on the right. It consists of two subscriptions with three resource groups in the first and one resource group in the second.

The first subscription is focused more on network resources like VNets and NSGs.

The second subscription is focused more on storage accounts, VMs, maintenance and Alerts. This environment contains many resources that are protected and monitored for optimal security

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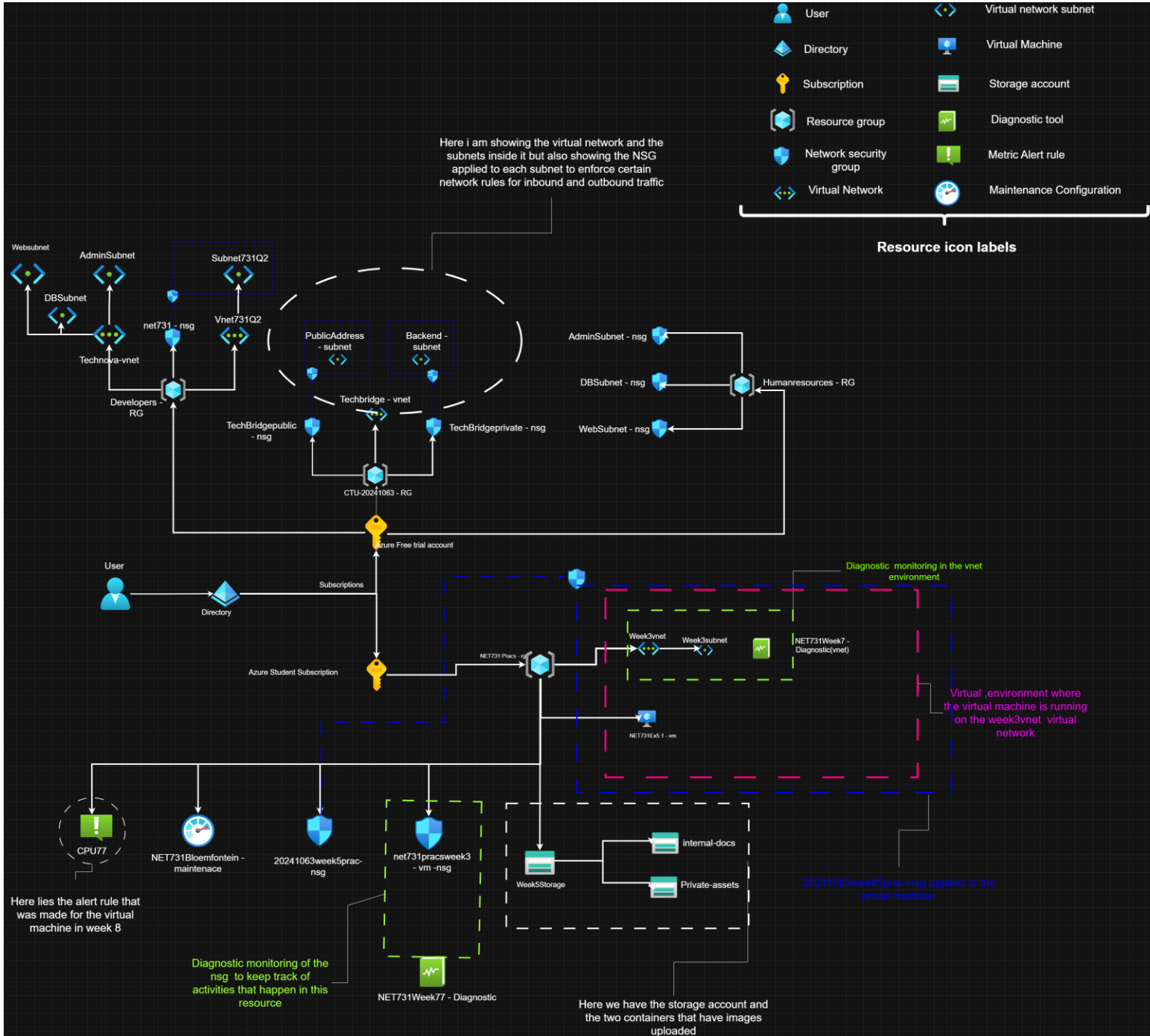
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Key Technical Decisions



net731pracsweek3-vm

The Version size of the VM I chose was a B2ls v2 due to its size because I deployed this virtual machine for general purpose use and acceptable ram size of 4GB and 2 CPU cores.



net731pracsweek3-vm-nsg

This NSG, I created for the virtual machine. The configuration I made a port 22 inbound rule so that remote access can be achieved.



CPU77

This is an alert I deployed specifically for the VM to get notified when CPU usage exceeds the limit is set, which is 77%.

Security and Monitoring



net731-nsg

subnet731Q2



Implemented an NSG to this specific subnet to allow traffic from the virtual machine only and the internet only. Allowing port 433 and 80



NET731Week7-Diagnostic(vnet)

week3subnet



week3vnet



Implemented a diagnostic tool in the virtual network to track network traffic and activities in the vnet.



NET731Week7-Diagnostic

Net731pracsweek3-vm-nsg



Implemented a diagnostic tool to track and monitor the activities in the NSG.

Cost Management

Service name ▾



Storage
\$2.14

Virtual Network
\$1.24

Virtual Machines
\$0.30

Azure Monitor
<\$0.01

Bandwidth
<\$0.01

Log Analytics
\$0.00

Location ▾



za north
\$3.67

global
\$0.00

Subscription ▾



Azure for Students
\$3.67

The total amount of money spent is 3 dollars 48 cents

Storage: **\$2.14**

Virtual machines : **\$0.30**

Bandwidth: **\$0.01**

Virtual Network: **\$1.24**

Azure Monitor: **\$0.01**

Log analytics: **\$0.00**

Location: **\$3.67**

Whole subscription/total: \$3.67

To manage cost I created and implemented a log analytics workspace to estimate where cost will be in the future judging by everyday usage.



Challenges and Solutions



Encountered a challenge when trying to deploy my virtual machine where I could not deploy a B1s size so I deployed a B2s v2 version instead.

Here, I encountered a challenge when creating a log analytics workspace is that I could not deploy it in my specific region to which I just deployed in another region



When creating a data collection rule for the VM the virtual machine had to be turned off for the resource to be properly deployed

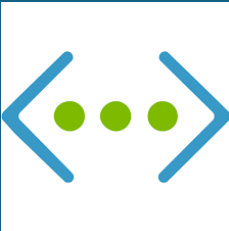
Lessons Learned



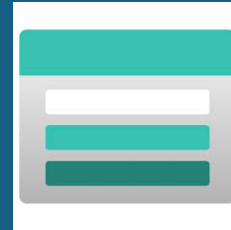
Multiple rules can be inserted into a NSG.



I learned that you can monitor specific network resources to track resource activity and monitor cost.



You can create a certain amount subnets in a virtual network



I learned how to create a storage account and insert Blob files



I learned that you can monitor cost for specific service in azure



Learned how to create an alert for my virtual machine

Conclusion

◆ Key take aways

- ◆ The smallest size VM that can be deployed is no longer a B1s, its B2s.
- ◆ You can set alerts based on the amount to of cost you accumulate per month
- ◆ You can schedule virtual machine update maintenance automatically
- ◆ SSH keys ae stored in the azure portal and on your PC
- ◆ You cant create certain resources with certain subscriptions
- ◆ VMs costs are measured per hour of use